Minimum Wages and Employment: A Replication of Card & Krueger (1994)

Individual Project — ECON6067 Quantitative Tools for Economic Analysis
Replication of: Card, D., & Krueger, A. B. (1994), *A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania*, AER 84(4): 772–793.

1. Research Question and Motivation

Does raising the minimum wage reduce employment? Competitive labor-market theory predicts that a binding wage floor reduces labor demand; monopsony and efficiency-wage accounts predict it need not. Card and Krueger (1994) exploit a natural experiment: on 1 April 1992 New Jersey (NJ) raised its minimum wage from $4.25 to $5.05, while neighboring eastern Pennsylvania (PA) kept $4.25. We compare fast-food restaurant employment before (Feb–Mar 1992) and after (Nov–Dec 1992) the hike using a difference-in-differences (DID) design.

2. Data, Sample Construction, and Variables

Data. Store-level survey of 410 Burger King, KFC, Roy Rogers, and Wendy's restaurants in NJ and eastern PA (`data/raw/public.csv`). Two waves; suffix 2 denotes the post-hike survey. Public replication data obtained from the Princeton Industrial Relations Section data archive (`irs.princeton.edu/publications/data-sets/minimum-wages-and-employment-chapter-4`, Krueger; archived at `arks.princeton.edu/ark:/88435/dsp0170795766s`), analyzable in Python, Stata, or R.

Outcome. Full-time-equivalent employment, `FTE = EMPFT + 0.5·EMPPT + NMGRS`. **Treatment.** `STATE`=1 (NJ); control = PA.

Sample. Raw 410 stores. We drop stores missing wave-1 or wave-2 FTE, and keep a balanced panel of stores observed in both waves: **384 stores (309 NJ, 75 PA)**. Ten stores closed or refused in wave 2; a robustness check re-includes permanent

closures as zero FTE. Dropped counts are logged in `code/01_make_dataset.py`.

First stage. Mean starting wage was ~$4.61 in both states in wave 1; by wave 2 it rose to $5.08 in NJ (at the new $5.05 floor) but stayed ~$4.62 in PA. The hike bit in NJ and not in PA.

3. Empirical Design

Two-period DID on the balanced panel:

$$\Delta FTE_i = \alpha + \beta \cdot NJ_i + \varepsilon_i$$

where ΔFTE = FTE(wave 2) − FTE(wave 1). β = (NJ mean ΔFTE) − (PA mean ΔFTE). We add chain fixed effects and, in a third specification, region and company-ownership controls. The key identifying assumption is parallel employment trends across the border absent the hike.

4. Core Replication Results

Table 3 — Mean FTE by state and wave (DID)

Group	Wave 1	Wave 2	Change	n
Pennsylvania (control)	23.38	21.10	−2.28	75
New Jersey (treatment)	20.43	20.90	+0.47	309
DID (NJ change − PA change)	**+2.75 FTE**			

The paper reports a DID of **+2.76 FTE**. Our replication matches to within 0.01. The wave-1/wave-2 means differ from the published table by ≈0.05–0.1 FTE (e.g., PA 23.38 vs 23.33, NJ 20.90 vs 21.03) because the paper computes means on its own sample of responding stores, whereas we use the balanced panel of stores observed in both waves and exclude closed/refused stores; the DID itself is robust to this choice. The sign is the opposite of the competitive prediction: NJ employment rose *relative to* PA after the wage hike.

Table 4 — DID regressions (ΔFTE ~ NJ)

Specification	DID coef (NJ)	Std. Error	t	n	R^2

(1) No controls	2.75	1.15	2.38	384	0.015
(2) Chain fixed effects	2.78	1.15	2.41	384	0.029
(3) Chain + region + co-owned	1.29	1.57	0.82	384	0.040

The estimate is stable (~2.8, significant at 5%) once chain composition is controlled. It shrinks and loses significance when fine region dummies are added, because the PA control group has only 75 stores split into two regions, leaving little within-region identifying variation. Robustness: counting permanently closed stores as zero wave-2 FTE leaves the DID at +2.65.

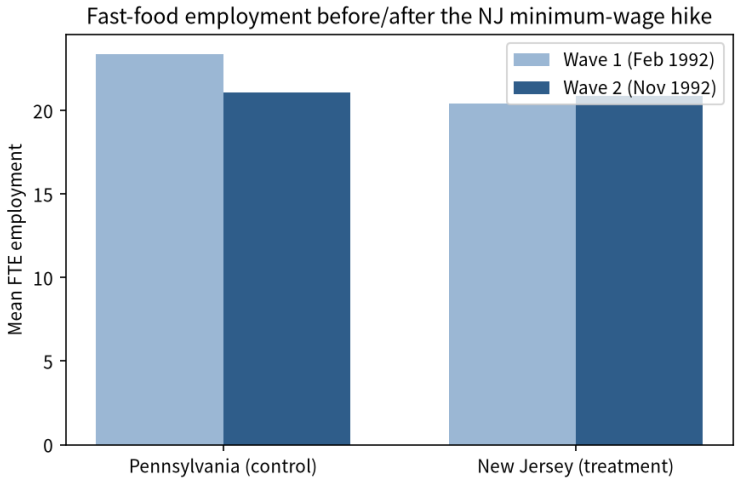


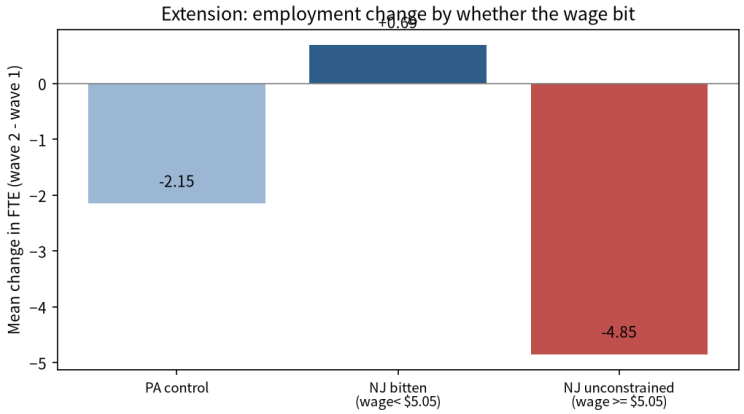
Figure 1. Mean FTE employment before and after the NJ minimum-wage hike.

5. Independent Extension: Did the Wage "Bite" Matter?

Question. The competitive model predicts the effect should be strongest where the new floor actually raised wages. We split NJ stores by wave-1 starting wage: *bitten* stores (wage < $5.05, forced to raise pay) vs. *unconstrained* stores (wage ≥ $5.05, already above the floor), and compare each to the PA control. The PA control here has n=72 (3 stores lack a wave-1 starting wage and cannot be classified as bitten or unconstrained).

Group	n	Mean ΔFTE	DID vs. PA
PA control	72	−2.15	—
NJ bitten (wage < $5.05)	268	+0.69	**+2.83 (t=2.48)**
NJ unconstrained (≥ $5.05)	25	−4.85	−2.70 (n.s.)

The positive NJ-vs-PA result is driven precisely by the stores whose wages had to rise: bitten stores gained a significant 2.83 FTE relative to PA. Unconstrained stores are too few (n=25) to read strongly. This reinforces the paper's conclusion: even where the minimum wage bit, employment did not fall.



Extension. Mean employment change by whether the new minimum wage bit.

6. Validation, Limitations, and Conclusion

Validation. We independently recomputed group means and the DID by hand; the wave-1/wave-2 means and the DID match the paper to ~0.01 FTE. The first-stage wage jump, balanced-panel count (384), and stability across chain controls were all checked. The region-control sensitivity and a collinearity issue (all PA stores fall in PA1/PA2, which is perfectly collinear with STATE) were diagnosed and documented.

Limitations. This is a border DID, not a fully identified causal effect: PA has only 75 stores; parallel-trends cannot be tested with two waves; adding fine region controls weakens the estimate; the unconstrained-NJ subsample is small.

Conclusion. We replicate Card & Krueger's headline finding: a minimum-wage increase was not associated with lower fast-food employment in NJ relative to PA (DID ≈ +2.75 FTE), and the effect is concentrated among the stores whose wages actually had to rise.

> **AI-use disclosure:** see `AI_USE_DISCLOSURE.md`. AI drafted the cleaning/regression code; every number was recomputed by hand and checked against the paper. Two AI errors were corrected: an FTE definition that omitted managers (changed to match the

> paper), and a rank-deficient region regression (fixed by dropping PA2).